EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 06

Basic Maths and Calculus

PW 04 A.P./G.P. leftover + full PW Class 05

By — Excellence Programme

9 September 2026

Topics to be covered

01 Last class in 10 minutes — trigonometry, max–min, phase and superposition formulae

02 Untaught PW Lecture 04 from slide 17 — complete A.P. and G.P.

03 Every PW Lecture 04 question after slide 17 — full working and answer

04 PW Class 05 — coordinates, distance and slope

05 Equation of straight line $y = mx + c$; slope and intercept

06 Draw lines and use them in velocity, ideal-gas and temperature graphs

Phase and superposition — formulae only

phase = complete angle inside sin or cos

phase difference: first make both sine or both cosine, then subtract

$$a\sin(\omega t) + b\cos(\omega t) = R\sin(\omega t+\varphi) \qquad R=\sqrt{(a^2+b^2)},\ \tan\varphi=b/a$$

$4\sin(\omega t) + 3\cos(\omega t) = 5\sin(\omega t + 37°)$

Trigonometry and max–min — formulae only

$180^\circ = \pi$ rad arc $= R\theta$ θ must be radian

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1$ for $\theta \le 5^\circ$, θ in radian

$a \sin\theta + b \cos\theta$ has maximum $\sqrt{(a^2+b^2)}$

PW Lecture 04 stopped before slide 17

Completed: phase, phasor and superposition.

Not completed: PW Lecture 04 slides 17 onward — A.P., G.P. and every question.

Today: A.P. → all A.P. questions → G.P. → all G.P. questions → PW Class 05 graphs

Do not treat A.P. and G.P. as recap. They are taught fully now.

Rough work / extra work

A.P. — equal difference every time

Arithmetic progression is a sequence in which the difference between consecutive terms is always constant.

a, a+d, a+2d, a+3d, ...

$$T_n = a + (n-1)d$$

$$S_n = n/2\,[2a + (n-1)d]$$

a = first term. d = next term − previous term. n = position.

A.P. example: 2, 5, 8, 11, 14, …

Find a, d, the 10th term and the sum of the first 10 terms.

$a=2 \qquad d=11-8=3$

$T_{10} = 2 + 9(3) = 29$

$S_{10} = 10/2\ (2+29) = 5\times31 = 155$

$T_{10}=29 \qquad S_{10}=155$

Question: 7, 11, 15, 19, ...

Find the 10th term and sum of the first 10 terms.

a=7 d=4

$T_{10} = 7 + 9(4) = 43$

$S_{10} = 10/2\ (7+43) = 250$

10th term=43 sum=250

Which term of 27, 24, 21, … is zero?

Which term of the A.P. 27, 24, 21, … is zero?

$a=27 \qquad d=-3 \qquad T_n=0$

$0 = 27 + (n-1)(-3) = 30-3n$

$3n=30 \qquad n=10$

The 10th term is zero.

Question: 28, 22, x, y, 4

28, 22, x, y, 4 is an A.P. Find x and y.

$a=28 \qquad d=22-28=-6$

$x=T_3=28+2(-6)=16$

$y=T_4=28+3(-6)=10$

$x=16 \qquad y=10$

Rough work / extra work

Sum of first n natural numbers

Find 1+2+3+...+n.

1. $n^2/2$
2. $n(n+1)/2$
3. $n(n-1)/2$
4. $n^2(n+1)/6$

a=1, d=1, last term=n **$S_n=n/2(1+n)$**

Option 2 **n(n+1)/2**

Physics question: 25 tuning forks

25 frequencies form an increasing A.P. Difference=10 Hz; 25th frequency=400 Hz. Find the first.

$T_{25} = a + 24d$

$400 = a + 24(10)$ $\quad a=400-240=160$

First tuning-fork frequency = 160 Hz

G.P. — equal ratio every time

Geometric progression is a sequence in which the ratio of consecutive terms is constant.

$a,\ ar,\ ar^2,\ ar^3,\ \ldots$

$$T_n = ar^{n-1}$$

$$S_\infty = a/(1-r), \text{ only if } |r|<1$$

r = any term ÷ previous term.

Which sequences are G.P.?

sequence	ratio	answer
2,4,8,12,24,48	ratio changes	not G.P.
2,4,8,16,32,64	2	G.P.
1,1/2,1/4,1/8,...	1/2	G.P., $S_\infty = 2$
1,−1/2,1/4,−1/8,...	−1/2	G.P., $S_\infty = 2/3$

A sequence is G.P. only if every consecutive ratio is the same.

Infinite G.P.: 1 + 1/3 + 1/9 + ...

Find 1 + 1/3 + 1/9 + 1/27 + ...

a=1 r=1/3 |r|<1

$S_\infty = a/(1-r) = 1/(1-1/3) = 1/(2/3)$

$S_\infty = 3/2$

Rough work / extra work

Potential at origin — alternating charges

+q, −q, +q, … are at x=1,2,4,8,… . Find potential at origin.

V = kq [1 − 1/2 + 1/4 − 1/8 + …]

a=1 r=−1/2 S∞=1/[1−(−1/2)]=2/3

V = 2kq/3

Infinite resistance network

Equivalent series blocks are R, R/2, R/4, R/8, … . Find R_AB.

$R_{AB} = R\,[1+1/2+1/4+1/8+\ldots]$

$R_{AB} = R \times 1/(1-1/2)$

$R_{AB} = 2R$

Force on q_0 at the origin

Charges q lie at x=1,2,4,8,… . Find force on q_0 at origin.

$F = kqq_0 [1/1^2 + 1/2^2 + 1/4^2 + \ldots]$

$F = kqq_0 [1 + 1/4 + 1/16 + \ldots] = kqq_0/(1-1/4)$

$|F| = 4kqq_0/3$ (toward $-x$ for like charges)

Fifth resistance when each is half

Five resistors are in series. Each resistance is half the previous. First is 10 Ω. Find the fifth.

$a = 10\ \Omega$ $r = 1/2$ $T_5 = ar^4$

$R_5 = 10(1/2)^4 = 10/16$

$R_5 = 5/8\ \Omega = 0.625\ \Omega$

A.P. and G.P. are now complete

A.P.: $T_n = a + (n-1)d$ $\quad$ $S_n = n/2[2a + (n-1)d]$

G.P.: $T_n = ar^{n-1}$ $\quad$ $S_\infty = a/(1-r)$, $|r| < 1$

A.P. → constant difference $\quad$ **G.P. → constant ratio**

Every PW Lecture 04 question after slide 17 is now answered. Start full PW Class 05.

Rough work / extra work

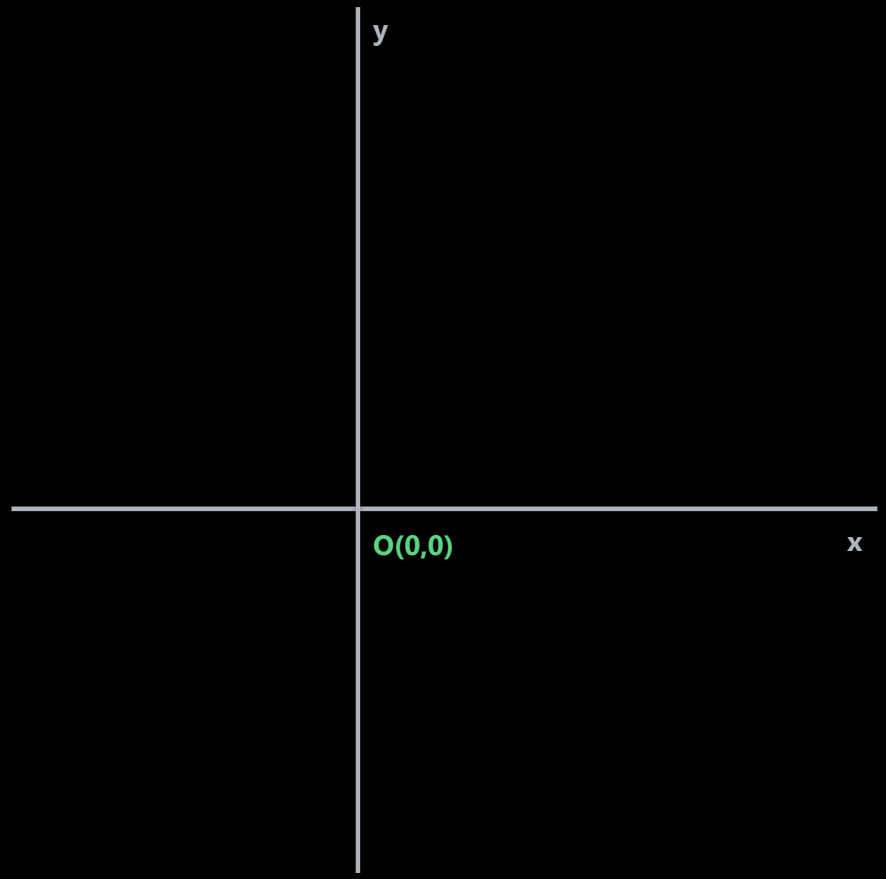
Graph — a picture of relation

A graph is a diagram in which a line or curve shows the relation between two physical quantities.

x-axis: horizontal **y-axis: vertical**

origin: O(0,0)

Always write the physical quantity and its unit on each axis.



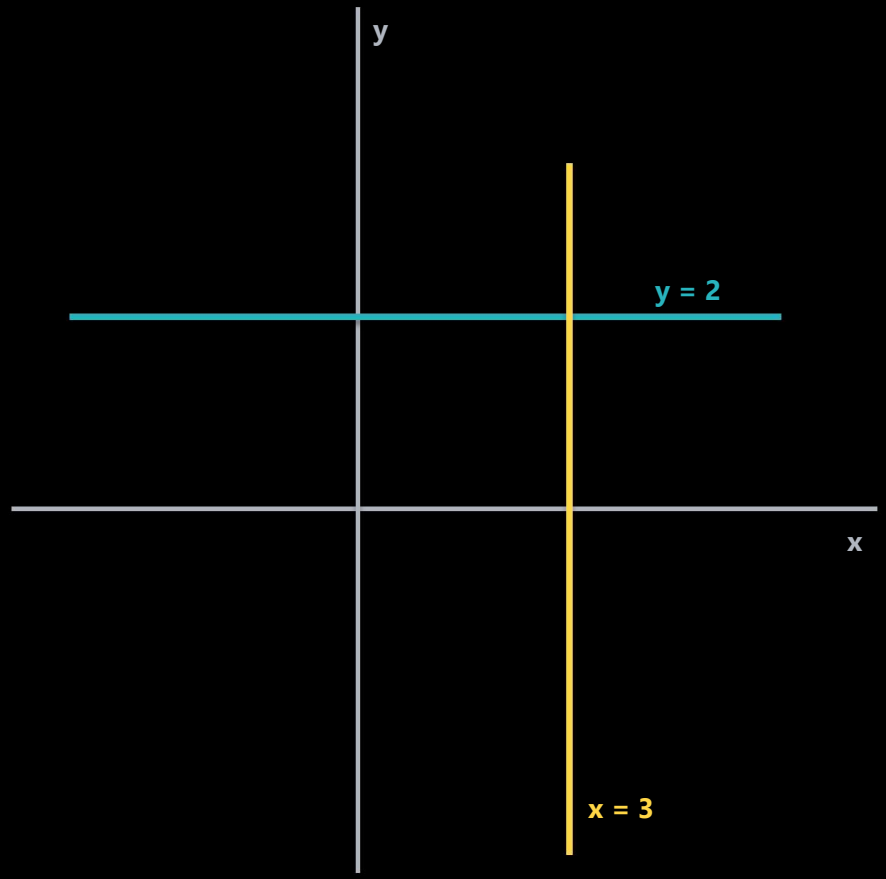
Constant lines — do not swap them

y = 2 → **horizontal line**

x changes but y stays 2.

x = 3 → **vertical line**

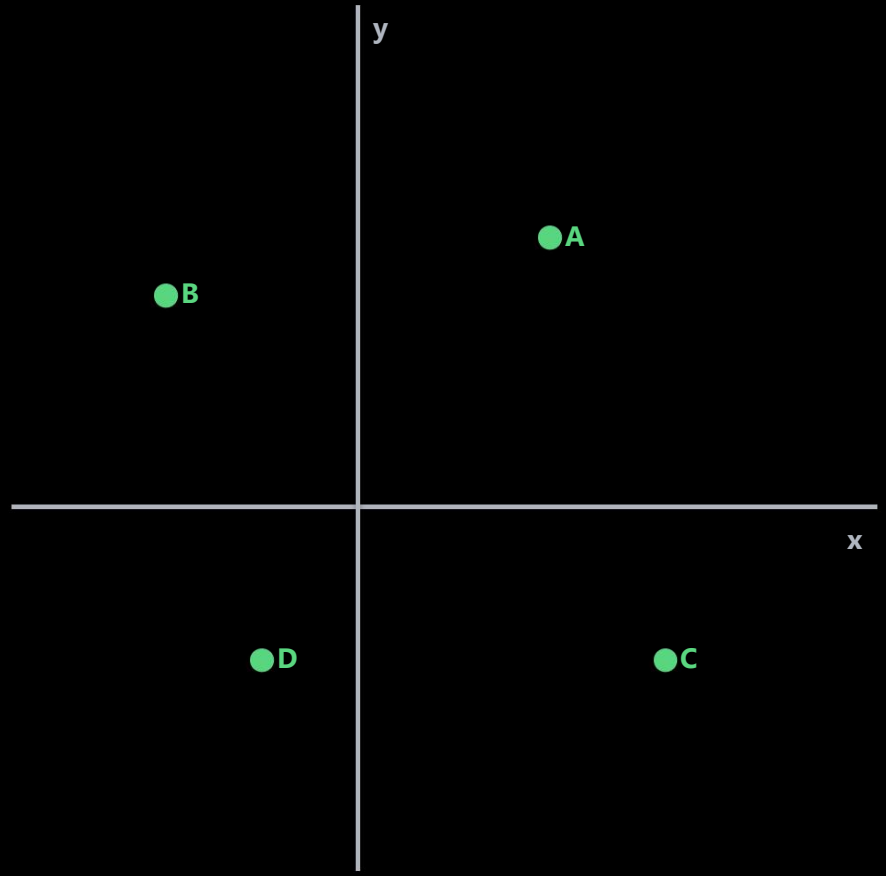
y changes but x stays 3.



Locate four points

Locate A(2,4), B(−2,3), C(4,−2), D(−1,−2).

point	x sign	y sign	quadrant
A(2,4)	+	+	I
B(−2,3)	−	+	II
C(4,−2)	+	−	IV
D(−1,−2)	−	−	III



Distance formula — two dimensions

For $P_1(x_1,y_1)$ and $P_2(x_2,y_2)$, make a right triangle using horizontal and vertical changes.

$\Delta x = x_2 - x_1$ $\Delta y = y_2 - y_1$

$$P_1P_2 = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

Keep each difference inside a bracket before squaring.

Distance in three dimensions

Find distance between P(2,−1,0) and Q(3,1,2).

$$PQ = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$$

$PQ = \sqrt{[(3-2)^2 + (1+1)^2 + (2-0)^2]}$

$PQ = \sqrt{(1 + 4 + 4)} = \sqrt{9} = 3$

Distance = 3 units

Rough work / extra work

Slope — rise divided by run

Slope tells how quickly y changes when x changes.

$$m = \text{slope} = \tan\theta = \Delta y/\Delta x = (y_2 - y_1)/(x_2 - x_1)$$

θ is the angle made by the line with the positive x-axis, measured anticlockwise.

A straight line has the same slope everywhere

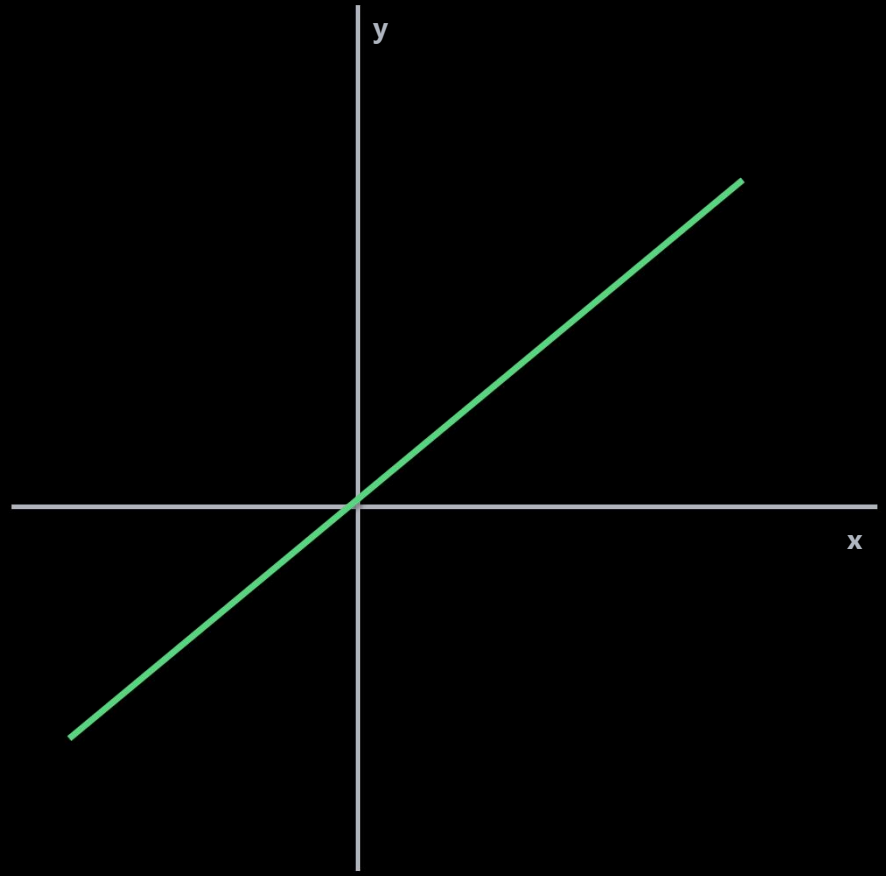
Choose any two points A and B on one straight line.

$$m_{AB} = (y_B - y_A)/(x_B - x_A)$$

Choose another pair C and D. The answer is unchanged.

$$m_{AB} = m_{CD}$$

This is special to a straight line. A curve generally has changing slope.

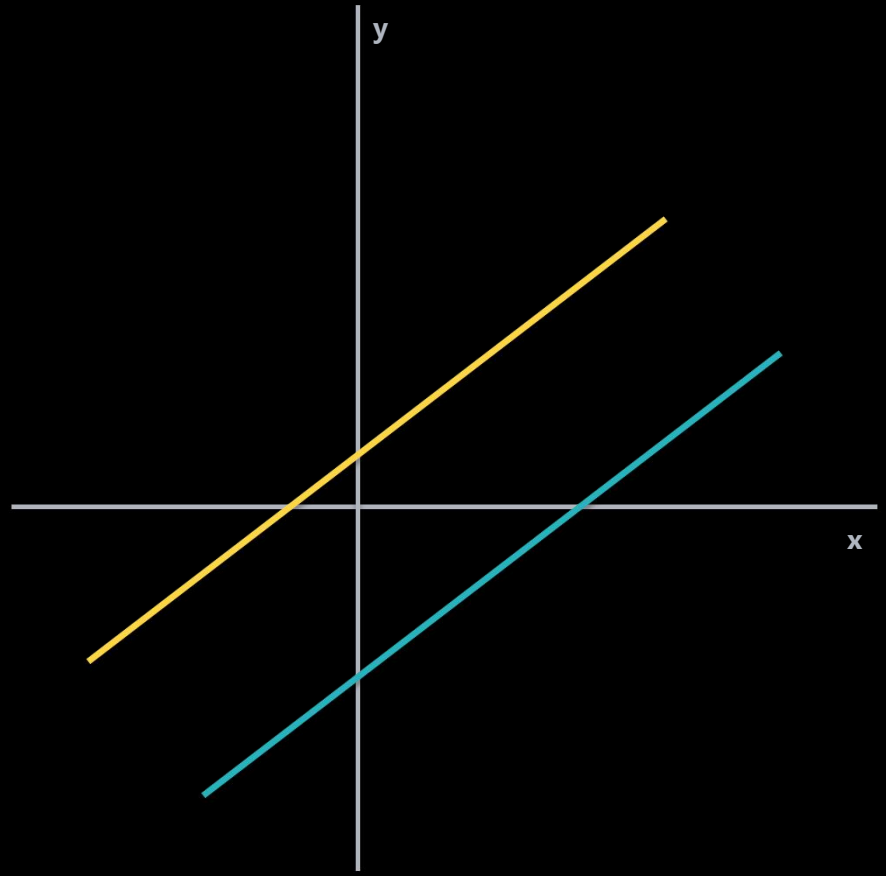


Parallel straight lines have equal slope

Parallel lines make the same angle θ with the positive x-axis.

$m_1 = \tan \theta$ $m_2 = \tan \theta$

parallel lines → $m_1 = m_2$



Slope values you must know

angle θ	slope tan θ	line feel
0°	0	horizontal
30°	1/√3	gentle rise
37°	3/4	rise
53°	4/3	steeper rise
90°	not defined / infinite	vertical
120°	−√3	falls
150°	−1/√3	gentle fall

Sign of slope from the angle

$0° < \theta < 90° \quad \rightarrow \quad m > 0$	$90° < \theta < 180° \quad \rightarrow \quad m < 0$

$\theta = 0° \rightarrow m = 0$ $\theta = 90° \rightarrow$ vertical slope is not defined

A rising line has positive slope. A falling line has negative slope.

Rough work / extra work

MR exam trap — slope is not the angle

"The angle made by a straight line with +x-axis is called slope." True or false?

angle = θ
slope = tan θ

False — slope is tan θ, not θ.

Ratio of slopes — 60° line and 30° line

Line A makes 60° and line B makes 30° with +x-axis. Find m_A/m_B.

m_A/m_B = tan60°/tan30°
m_A/m_B = √3 ÷ (1/√3) = 3

m_A : m_B = 3 : 1

Ratio of slopes — 30° line and 60° line

Now line A makes 30° and line B makes 60°. Find m_A/m_B.

$m_A/m_B = \tan30°/\tan60°$

$m_A/m_B = (1/\sqrt{3}) \div \sqrt{3} = 1/3$

$m_A : m_B = 1 : 3$

Equation of a straight line

Every non-vertical straight line can be written in slope-intercept form.

$$y = mx + c$$

m = slope **c = y-intercept = value of y when x=0**

m and c are fixed for one straight line. x and y change from point to point.

Read slope and y-intercept directly

First arrange the equation so y is alone. Then compare term by term.

$y = (4/3)x + 4 \quad \rightarrow \quad m = 4/3,\ c = 4$

$y = -\sqrt{3}\,x - 4 \quad \rightarrow \quad m = -\sqrt{3},\ c = -4$

The constant c is where the line cuts the y-axis, not the x-axis.

Rough work / extra work

PW example: c = 4 and θ = 53°

A line cuts the y-axis at 4 and makes 53° with +x-axis. Write its equation and find y at x=3.

m = tan53° = 4/3 c = 4

y = (4/3)x + 4

at x=3: y = (4/3)(3) + 4 = 8

Equation: y=(4/3)x+4 y(3)=8

PW example: $c = -2$ and $\theta = 30^\circ$

A line cuts y-axis at -2 and makes 30°. Write its equation. Find y at $x=2\sqrt{3}$.

$m = \tan 30^\circ = 1/\sqrt{3}$ $\quad$ $c = -2$

$y = x/\sqrt{3} - 2$

$y(2\sqrt{3}) = 2\sqrt{3}/\sqrt{3} - 2 = 0$

$y = x/\sqrt{3} - 2$ $\quad$ $y(2\sqrt{3})=0$

Signs of m and c — four quick pictures

line behaviour	m	y-axis crossing	c
rises, crosses above origin	+	above	+
falls, crosses above origin	−	above	+
rises, crosses below origin	+	below	−
falls, crosses below origin	−	below	−

Do not use where the line cuts the x-axis to decide c.

PW example: c = 5, θ = 150°, x = −4

A line has y-intercept 5 and angle 150°. Write its equation and find y at x=−4.

$m = \tan 150° = -1/\sqrt{3}$ $c = 5$

$y = -x/\sqrt{3} + 5$

$y(-4) = 4/\sqrt{3} + 5$

$y = -x/\sqrt{3} + 5$ $y(-4)=5+4/\sqrt{3}$

PW example: $c = -4$ and $\theta = 120^\circ$

A line cuts the y-axis at -4 and makes angle 120°. Write its equation.

$m = \tan 120^\circ = -\sqrt{3}$ $\quad$ $c = -4$

$y = mx + c$

$$y = -\sqrt{3}\,x - 4$$

Rough work / extra work

Horizontal and vertical line equations

horizontal through (0,4): m=0, c=4 → y=4

vertical through (3,0): x=3

A vertical line cannot be written as y=mx+c because its slope is not defined.

PW intercept example: x-cut = −3, y-cut = 4

A line crosses the x-axis at −3 and y-axis at 4. Write its equation.

c = 4 m = rise/run = 4/3

y = (4/3)x + 4

PW intercept example: x-cut = 4, y-cut = 3

A line crosses the x-axis at 4 and y-axis at 3. Write its equation.

$c = 3$ $\quad$ $m = (0-3)/(4-0) = -3/4$

$$y = -(3/4)x + 3$$

Draw y = √3x + 4 and y = x

equation	m	c	angle
y=√3x+4	√3	4	60°
y=x	1	0	45°

Method: mark (0,c), use the angle or rise/run, then extend the straight line.

Draw y = −x and y = (4/3)x − 5

equation	m	c	line
y=−x	−1	0	falls through origin
y=(4/3)x−5	4/3	−5	rises from below origin

Negative c does not automatically mean negative slope. m and c tell different things.

Rough work / extra work

Find intercepts before drawing

Draw $y=2x-1$ and $y=-2x+4$.

equation	y-intercept	x-intercept	slope
$y=2x-1$	-1	$1/2$	$+2$
$y=-2x+4$	4	2	-2

For x-intercept, put y=0. For y-intercept, put x=0.

Convert to y = mx + c first

Write each in slope-intercept form.

(1) $\frac{x}{3} + \frac{y}{4} + 1 = 0$

$\frac{y}{4} = -\frac{x}{3} - 1$ $\quad$ $y = -\frac{4}{3}x - 4$

$m = -\frac{4}{3}$, $c = -4$

(2) $(y-3)+(x-4)+5=0 \rightarrow x+y-2=0 \rightarrow y=-x+2$

$m=-1$, $c=2$

Compare two lines without drawing

Two lines both rise. Line 1 crosses y-axis below origin; line 2 crosses above. State signs of m and c.

line	slope m	intercept c
1	positive	negative
2	positive	positive

Both m are +ve. Only line 2 has +ve c.

Y-intercept 4 and point (2,6)

Draw a line with y-intercept 4 passing through (2,6). Find its equation.

$c=4$ $6 = m(2)+4$ $2m=2$ $m=1$

$y = x + 4$

Point (2,3) and slope 1

Draw a line passing through (2,3) with slope 1. Find its equation.

$y=mx+c$ $3=1(2)+c$ $c=1$

$y = x + 1$

Rough work / extra work

Line through (3,4) and (2,6)

Find equation of the straight line through (3,4) and (2,6).

$m = (6-4)/(2-3) = 2/(-1) = -2$

$4 = -2(3)+c \qquad c=10$

$$y = -2x + 10$$

Physics graph: v = 2t − 4

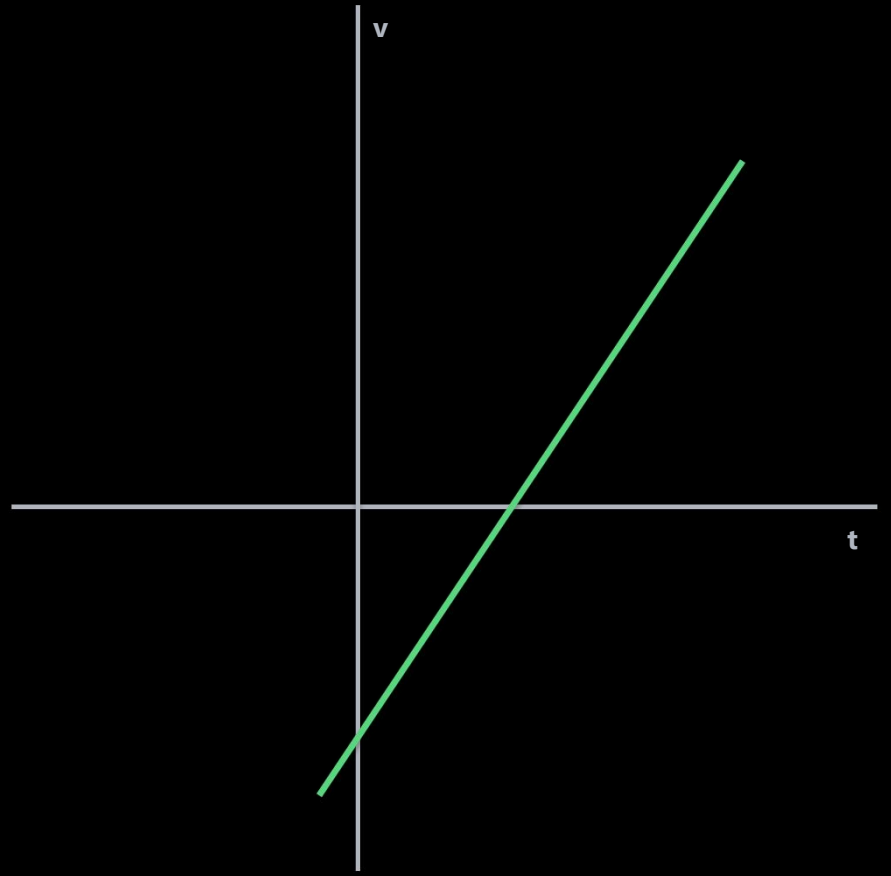
Velocity is v=2t−4. Draw the velocity–time graph.

compare v=mt+c

slope = 2 v-intercept = −4

v=0 → 2t−4=0 → t=2 s

The line crosses the time-axis at 2 s.



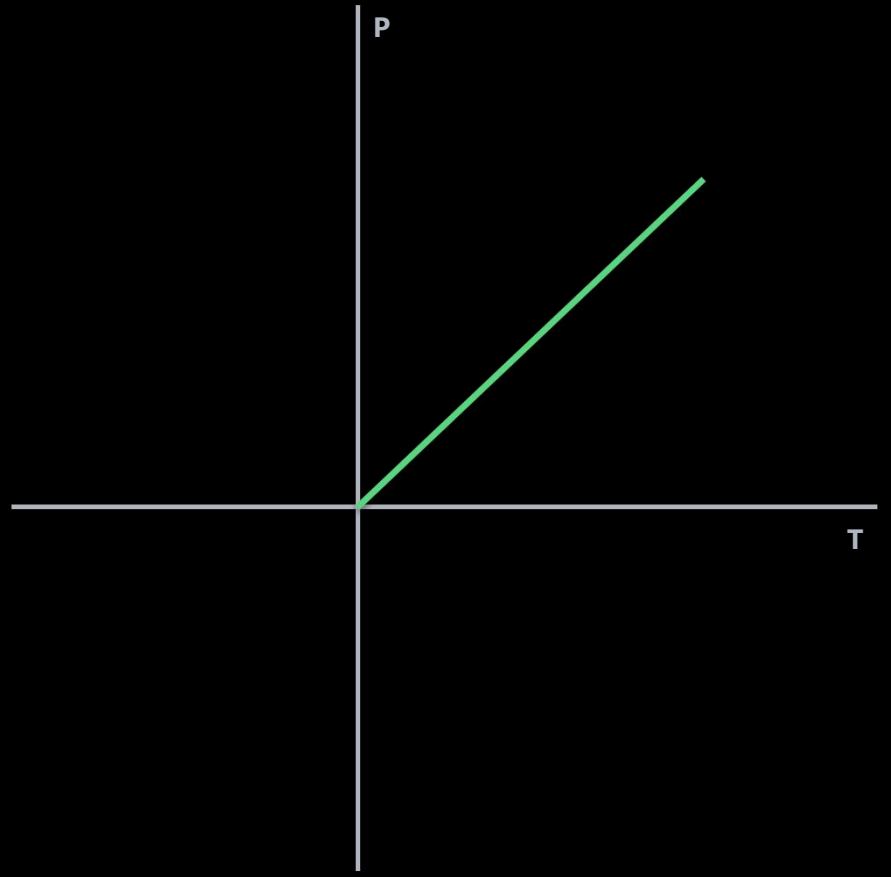
Ideal gas: P versus T at constant volume

If PV=nRT and volume is constant, draw P versus T.

P = (nR/V) T

m = nR/V > 0 c=0

So it is a rising straight line through the origin.



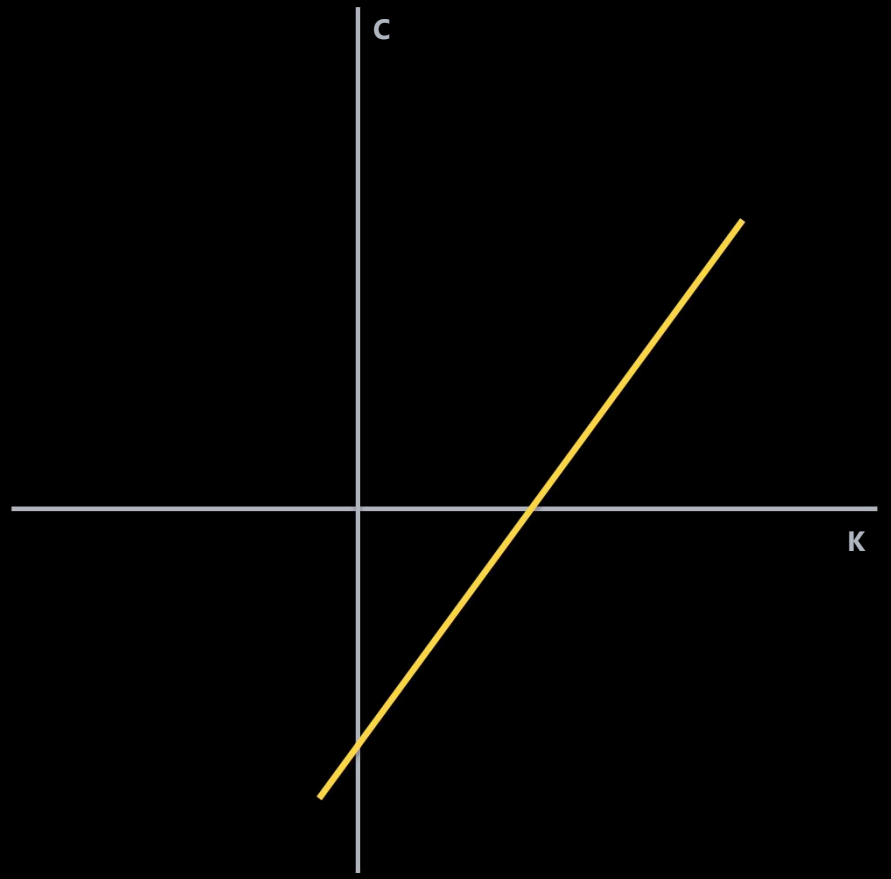
Celsius and kelvin — a straight line

Given C/100 = (K−273)/100, draw C versus K.

C = K − 273

slope = 1 C-intercept = −273

C=0 at K=273



PW Class 05 homework — intercept question

Find the equation of the line whose x-intercept is +4 and y-intercept is +5.

1. $5x + 4y + 20 = 0$
2. $4x + 5y - 20 = 0$
3. $4y - 5x = -20$
4. $4x + 5y + 20 = 0$

Tonight: solve by using the two points (4,0) and (0,5).

Rough work / extra work

PW Class 05 homework — draw four graphs

Draw graph between y and x for each equation.

(i) $y = \sqrt{3}x + 4$

(ii) $y = -x$

(iii) $y = |x|$

(iv) $x/3 + y/4 + 1 = 0$

For every graph: first write m and c, then mark two points.

After class

Daily test before the next lecture.
Copy every solved PW Class 05 question once in your notes.

Solve the intercept MCQ and four graph questions on rough pages.

Coordinates, distance and slope

$$\text{distance}_2D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

$$\text{distance}_3D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$$

$$m = \tan\theta = \Delta y/\Delta x$$

rising → m positive falling → m negative horizontal → 0 vertical → undefined

Straight-line lock-lines

$$y = mx + c$$

m = slope **c = y-intercept**

x-intercept: put y=0 **y-intercept: put x=0**

parallel lines have equal slope

Slope is tanθ, not θ. A vertical line is x=constant.